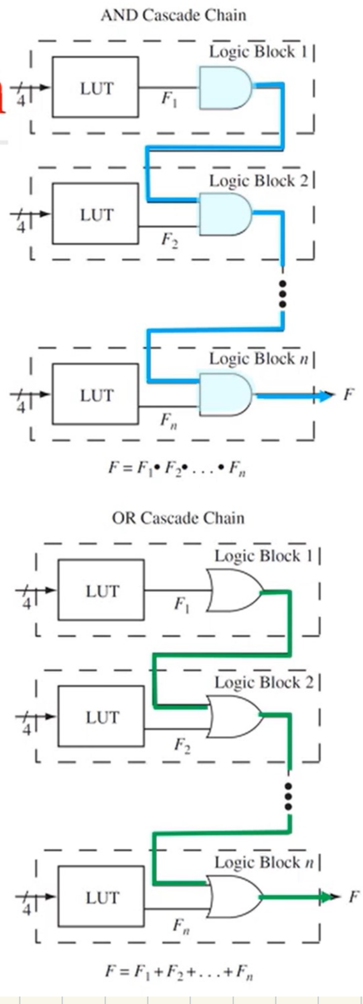
6.4 Cascade Chains in FPGAs

Cascade Chains in FPGAs

- ***Cascade chain***: cascading **outputs** from FPGA blocks in series
- Common types of cascading:
  - ***AND cascade chain***: useful for creating **wide-AND gates**
  - ***OR cascade chain***: useful for creating **wide-OR gates**
  - ***Register cascade chain***

AND/OR Cascade Chain

- AND/OR cascade chain:
  - Instead of using separate function generators to perform AND or OR functions of logic block outputs, the output from one logic block can be directly fed to the cascade circuitry to create AND or OR functions of the logic block outputs.
  - In LUT-based FPGAs, these types of cascade chains may be called ***LUT chains***.



Example: AND Cascade Chain

- Assumption: Each logic block has one LUT4.
- How many logic blocks w/ ***LUT4s*** will be needed to create a ***32-variable AND*** function w/ and w/o the **AND cascade chain**?

<Ans.>

W/o the AND cascade chain:

W/ the AND cascade chain:

Example: AND Cascade Chain

$A_0\ A_1\ A_2\ A_3\quad A_4\ A_5\ A_6\ A_7\ \ldots\ A_{28}\ A_{29}\ A_{30}\ A_{31}$ → 8

× → 2

× → 1

- Assumption
- How many l
needed to cr
w/ and w/o the **AND cascade chain**?

<Ans.>

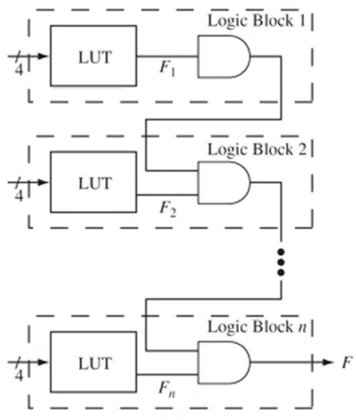
W/o the AND cascade chain:

11 logic blocks will be needed.

W/ the AND cascade chain:

8 logic blocks will be needed.

AND Cascade Chain



$F = F_1 \bullet F_2 \bullet \ldots \bullet F_n$

Register Chain

- Register chain:
  - There is a separate register chain input to the flip-flop.
  - It is possible to use the LUT for other logic functions and route their outputs using the combinational outputs X and Y,

  ⇒ Potentially increase the utilization of the FPGA blocks.